

Muriwai solar data — cleaned for the pilot

Cleaned power/energy data for the Muriwai 5kW + Batteries system, converted from the raw Solarman Excel exports into tidy CSVs. Input for the database and dashboard.

Files

File	What it is	Use it for
muriwai_readings_5min_clean.csv	Every raw reading, 5-min intervals, cleaned & standardised	Loading into the database at full resolution
muriwai_standard_30min.csv	The four project metrics aggregated to 30-min intervals	The "standard format" all sites convert into; feeds the dashboard

The four metrics Trust Tairāwhiti asked for

1. **Solar generation** — generation_kwh
2. **Electricity consumption** — consumption_kwh
3. **Grid import** — grid_import_kwh (power bought)
4. **Grid export** — grid_export_kwh (surplus sold)

Interval: 15–30 min (this file uses 30 min; change the resample step for 15).

Column reference — muriwai_readings_5min_clean.csv

Column	Unit	Meaning
timestamp	ISO 8601, NZ (UTC+12)	Reading time, every 5 min
site	text	Plant name (the site key when multiple marae are added)
generation_w	W	Instantaneous solar power
consumption_w	W	Instantaneous site load
grid_import_w	W	Power bought from grid
grid_export_w	W	Surplus exported to grid
battery_soc_pct	%	Battery state of charge
*_kwh	kWh	Energy in that 5-min interval (= W × 5/60 ÷ 1000)

Standard file columns

site, timestamp, generation_kwh, consumption_kwh, grid_import_kwh, grid_export_kwh — each row is the SUM of the 5-min energy values in that 30-minute bucket.

Important quirks (already handled here)

FEED-IN was stored **negative** in the raw export — flipped to a positive `grid_export` .

W → kWh energy = power(W) × interval(hours) ÷ 1000. Interval/daily totals are the **SUM** of these, not an average (this is how Solarman derives its own daily figures).

GAPS the raw exports drop some 5-min intervals (a full day should have 288), so export totals can be marginally low. The database should tolerate gaps and ideally flag them.

Source spans **1–14 Sep 2026** and **17 Sep 2026** (15–16 Sep not exported).

Cost variable (editable in the dashboard)

Let a user enter an **import rate** and **export rate** (\$/kWh):

1. cost of grid power = `grid_import_kwh × import_rate`
2. value of exports = `grid_export_kwh × export_rate`
3. avoided-cost value of solar used on-site = `(generation_kwh - grid_export_kwh) × import_rate`

Rates are illustrative until the marae's real tariff is confirmed.

Suggested database shape (single table, per-site, agnostic)

```
readings (  
  site          TEXT,  
  timestamp     DATETIME,  -- interval start  
  generation_kwh REAL,  
  consumption_kwh REAL,  
  grid_import_kwh REAL,  
  grid_export_kwh REAL,  
  PRIMARY KEY (site, timestamp)  
)
```

Every supplier's export is converted into THIS shape by a small per-supplier importer, so the database and dashboard never change when a new marae/brand is added. Muriwai/Solarman is the first importer.